Documents for Go2Joy · 2026-10-27

SharpTalk for Go2Joy

`HA sign-in integration spec v1.4`

Sign-in integration spec v1.4 for the Go2Joy HA dev team (2026-10-24, doc revision 10/27). Done means every item in the checklist at the top passes. Schedule and owners: work request. For printing or sending: PDF

> When an HA staff member opens the widget while signed in, **the HA server signs the staff and hotel data**, and HA FE passes that package to the widget **unchanged**. SharpTalk verifies the signature, registers the hotel as a business customer and the staff member as a contact, continues conversations per hotel account, and routes to the team for the hotel's region. **In v1.4, HA does not calculate the region.** HA sends only the province number (`province_sn`). SharpTalk sets the name and region from the Go2Joy province list.

Item	Details
To	Go2Joy Hotel Admin (HA) dev team (backend, FE) · Go2Joy PM
From	SharpTalk (Amoeba)
Date / version	2026-10-24 · **v1.4** · doc revision 2026-10-27 (wire format unchanged; see "Doc revision 10/27") · v1.3 2026-10-22 and v1.0 PDF 2026-10-20 are obsolete
Environments	HA UAT `https://go2joy-ha-uat.go2joy.io` ↔ SharpTalk staging `https://shoptalk.amoeba.site` · Production is the Go2Joy server `https://sharptalk.go2joy.vn` (install pending)
Related docs	Work request (WR numbers, schedule) · v1.3 (previous spec, obsolete)
Sample code	sharptalk_identity.php · vectors.php (runs the test vectors)

✅ What HA must do — checklist

This checklist is the core of this document. An item is done when its check passes.

> **Progress (HA UAT checked 10/27)**: done **B5 · B6 (UAT)**. Everything else has not started yet. B4 is partial (`logout` is called on the sign-out button and on token expiry; sign-out in another tab is not handled). B3: `identify` is called once at start-up, not on hotel change.

A. HA backend (WR-01 · WR-02) — Tue 10/27

- ☐ **A1. Keep the signing key (embed secret) only in HA server environment variables.** Do not put it in FE, the browser, the app, logs or documents. SharpTalk sends the staging value over a secure channel (on the day WR-02 is decided). Check: the secret string appears nowhere in the repository or bundles.
- ☐ **A2. Create an endpoint that can be called only with a signed-in session** (e.g. `GET /api/v1/staff/sharptalkIdentity`). Check: a call without sign-in returns 401.
- ☐ **A3. Read values from the HA session and DB.** Do not build them from values sent by the browser.
- ☐ **A4.** `userId = "ha:" + userSn` **.** Use the staff number, which never changes. Do not use the login ID (`go2joysupport` etc.).
- ☐ **A5. Exactly 4 attributes**: `hotel_id` · `hotel_name` · `province_sn` · `role` .
  - **Do not send keys outside this list**, such as `hotel_code` · `hotel_group_id` · `user_id` · `user_sn` · `district` (they are left out of the signature calculation, so you get 401).
  - Do not send `region` · `province` (name) either. During the transition period this is not 401, but `province_sn` takes priority. After the transition it becomes 401.
- ☐ **A6.** `province_sn` **= the Go2Joy** `provinceSn` **of the selected hotel** (Province Feed `sn` , numeric string, e.g. Hồ Chí Minh = `"1"`).
- ☐ **A7.** `role` **= the HA role code** (a value that does not depend on the UI language). Send SharpTalk the **code table** (code · Vietnamese/English display name).
- ☐ **A8. When "All hotels" is selected**, leave out `hotel_id` · `hotel_name` · `province_sn` and sign only `role` .
- ☐ **A9. Follow the normalization order**: **NFC → collapse whitespace (incl. NBSP) → trim → 120 chars (** `mb_substr` **)**. Requires the PHP `intl` · `mbstring`

extensions.

- [ ] **A10. Leave out empty values entirely (drop the field)**. This applies to attributes and to `name` · `email` · `phone`. **Also drop an email that is not a valid address**: an empty or malformed email returns 400 and the whole sign-in fails (the staff member stays a guest). The same happens for `userId` over 120 chars, `name` over 200 or `phone` over 40.
- [ ] **A11.** `hash = hex(HMAC-SHA256(secret, signing string))`. The signing string uses the format in 5.1 (keys in alphabetical order).
- [ ] **A12. Test vectors T1 ~ T6 all match** (5.4). Send the results to SharpTalk.
- [ ] **A13. Return the response package over HTTPS only to the signed-in session. Do not log it.**

B. HA FE (WR-04) — Wed 10/28

- [ ] **B1. Do not build the package in the browser. Delete** the current `sharptalkSignIn` (built from profile values) and the initial value `hash: "testing"`.
- [ ] **B2. Take the A2 response and pass it to** `SharpTalk.identify(response)` **unchanged** (do not add or change keys).
- [ ] **B3. Call again on**: sign-in confirmed · page reload · **hotel selection change (incl. All hotels)** · HA language change.
- [ ] **B4. Call** `SharpTalk.logout()` **on**: sign-out button · **automatic sign-out on session/token expiry** · right before another account signs in · sign-out in another tab (`storage` event etc.).
- [x] **B5.** `locale`: same as now, `localStorage.lang` → `vi` / `en`. Call `setLocale` when it changes. *(checked 10/27: done)*
- [x] **B6. Allow the SharpTalk domain in CSP (** `script-src` · `frame-src` · `connect-src` **) and CORS.** *(checked 10/27: done on UAT — no CSP restriction, the widget loads normally. Production is D3)*
- [ ] **B7. Put the 3 SharpTalk domain uses** (`embed.js` src · `widgetUrl` · `ivy:event` origin check) **behind one environment setting** (so you change it once when moving to production). Check (all of B): DevTools › Network `POST …/public/embed/identify` **200 with** `"authenticated": true`, and the console shows the session as **"verified"** (not "Unverified"). **Not proof on its own**: while staging still runs the temporary unsigned mode (§ 7), a wrong signature ALSO returns 200, `"consentState": "granted"` and `identified {ok: true}`.

C. Joint UAT check (WR-05) — 10/29 ~ 10/30

- ☐ **C1. All items in the section 8 checklist pass** (together with SharpTalk). § 8 is the only UAT list; it runs after SharpTalk ends the temporary unsigned mode (§ 7).
- ☐ **C2. Test accounts**: 1 `Quản lý` (multiple hotels) + 1 `Lễ tân` (1 hotel), including hotels in the north and the south (WR-14).

D. Production deploy (WR-06) — before production UAT

- ☐ **D1. Apply the production embed secret.** It differs from staging (SharpTalk sends it over a secure channel).
- ☐ **D2. Switch to the production domain**: `https://sharptalk.go2joy.vn/widget/embed.js` · `widgetUrl` · origin check.
- ☐ **D3. Apply production CSP and CORS. Share the HA deploy schedule in advance.** Check: identify returns 200 with `"authenticated": true` on production HA. A package made with the staging secret returns 401 on production.

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v1.4 changes

Item	v1.0 (10/20 PDF)	v1.3	v1.4
`userId`	Login ID	`"ha:" + userSn`	Same as v1.3
Signed attribute keys	`hotel_id` · `hotel_name` · `province` · `region` · `role`	Same	**`hotel_id`** · **`hotel_name`** · **`province_sn`** · **`role`**
Province	Province name	Province name	**`province_sn`** = Go2Joy `provinceSn`
Region	Calculated by HA (3 values)	Calculated by HA (5 codes)	**Not sent**. Calculated by SharpTalk
Role	Display name (`Lễ tân`)	Display name	**Role code**
Normalization	trim · 120 chars	+ NFC	**NFC → collapse whitespace → trim → 120 chars**
"All hotels"	Not defined	Call again only	Drop the 3 hotel keys, sign `role` only

Item	v1.0 (10/20 PDF)	v1.3	v1.4
Empty values	Empty attributes · email left out	Same	Empty `name` · `email` · `phone` also left out
Test vectors	T1 · T2	T1 ~ T5	**T1 ~ T6** (previous vectors no longer used)

Why the change

- If both HA and SharpTalk keep a region table, they drift apart. The single source is the Province Feed of the Go2Joy Partner API.
- Province names and role display names change with the HA language (vi/en), so the signed values could differ. Numbers and codes do not change.

Transition period: For now, SharpTalk still accepts the v1.3 format (`province` · `region` keys). If `province_sn` is also sent, **the values set from `province_sn` take priority**. Support for the v1.3 keys ends in a separate release after HA passes UAT on v1.4; the date is announced in advance.

Doc revision 10/27 (no change to what HA sends)

#	Where	Change
1	B check · § 5.5 · § 7 · § 8	`ok: true` is not proof while staging accepts unsigned sign-ins. Proof = `"authenticated": true` + console "verified". SharpTalk ends the temporary mode as soon as a verified identify is seen on HA UAT, **before** the joint UAT
2	§ 1 region table · § 8	No province is in region 3 (Central) today; central-coast provinces are region 5, Thừa Thiên Huế is region 4. UAT uses the 4 regions that exist
3	§ 1 Continuity · § 6	Per-hotel continuity described exactly: the widget shows each staff member's own conversations plus a count of the hotel's
4	A10 · § 5.5	400 also for a malformed email and over-long `userId` / `name` / `phone`; the whole sign-in fails
5	§ 7	Status updated: v1.4 support is done on staging (10/24)
6	§ 8	One UAT list for HA and SharpTalk (merged with work request § 6.1)
7	B5 · B6	Done on HA UAT (10/27)

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1. What SharpTalk does

Step	Details
Verify	Builds the signing string from the received values as-is and compares the HMAC. On mismatch: 401 E5048 (stays guest)
Province · region	`province_sn` → fills **province name and region** from the Go2Joy province list (80 entries, Partner API Province Feed, refreshed daily). A number not in the list is saved without a region and goes to the default assignee
Register	Hotel = business customer (`hotel_id`), staff = contact (`userId`), role recorded
Continuity	Continues per hotel account: the AI sees the hotel's previous conversation and the same agent is preferred for 24 h. In the widget each staff member sees their own earlier conversations plus a count of the hotel's (work request WR-18)
Routing	Consult category chosen by the customer × hotel region → team × work label (rules managed by Go2Joy in the console)

Region table (Province Feed `region`)

No.	Code	Name
1	`south`	South
2	`north`	North
3	`central`	Central — **no province in the current feed**. Central-coast provinces (Đà Nẵng, Quảng Nam, Bình Định, Khánh Hòa, Gia Lai …) are 5, Thừa Thiên Huế is 4
4	`north_center`	North Central
5	`south_center`	South Center
10	`region_10`	Not documented (17 cities of Manila). To be fixed after the reply to work request WR-27

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2. Data sent (v1.4)

Field	Signed	Value	HA source (example)	Example
`userId`	Yes	`"ha:" + userSn`	Staff number in HA DB	`ha:1001`
`attributes.hotel_id`	Yes	Selected hotel `hotelSn` (string)	Selected hotel in the session	`"1721"`
`attributes.hotel_name`	Yes	Hotel name	Hotel in HA DB	`A In Hotel Del Luna`
`attributes.province_sn`	Yes	`provinceSn` of the hotel's province (string)	Hotel in HA DB → provinceSn	`"1"`

Field	Signed	Value	HA source (example)	Example
`attributes.role`	Yes	HA role code	Role key in HA DB	`RECEPTIONIST` (example. Real codes come with the A7 reply)
`name` · `email` · `phone`	No	For display. Drop the field when empty	Profile	`G2J SUPPORT`

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3. Request flow

```
HA FE ──(signed-in session)──▶ HA server  GET /api/v1/staff/sharptalkIdentity
                          └ values from session/DB → normalize → sign → return packa
ge
HA FE ── SharpTalk.identify(package unchanged) ──▶ widget iframe
widget ── POST /api/v1/public/embed/identify ──▶ SharpTalk
                          └ verify signature → province_sn → province name/region →
business customer/contact → link conversation
widget ── ivy:event { event: "identified", ok } ──▶ HA FE (optional: log)
```

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4. Signing string

```
no attributes   : userId
with attributes : userId + "\n" + key=value + "\n" + key=value ...   (keys in alpha
betical order)

hash = hex( HMAC-SHA256( key = SHARPTALK_EMBED_SECRET, message = signing string (UT
F-8) ) )
       → 64 lowercase hex chars

Example (T2):
ha:1001
hotel_id=1721
hotel_name=A In Hotel Del Luna
province_sn=1
role=RECEPTIONIST
```

- Key order: `hotel_id` → `hotel_name` → `province_sn` → `role`.
- The values you send must be **exactly the same** as the values you signed (including spaces, letter case and tone marks).

5. Implementation

5.1 HA server (PHP) — sharptalk_identity.php

```
<?php
// SharpTalk identify package — HA server (spec v1.4). Requires PHP intl + mbstrin
g.

/** NFC → collapse whitespace (incl. NBSP) → trim → 120 chars. Order matters. */
function sharptalk_clean($value): string {
    $v = Normalizer::normalize((string) $value, Normalizer::FORM_C);
    if ($v === false) { $v = (string) $value; }
    $v = preg_replace('/[\s\x{00A0}\x{2000}-\x{200B}\x{3000}]+/u', ' ', $v);
    $v = trim($v);
    return mb_substr($v, 0, 120, 'UTF-8');
}

/**
 * @param string     $userSn   staff number from the HA DB (never the login ID)
 * @param array|null $hotel    ['sn' => …, 'name' => …, 'provinceSn' => …] or null
for "all hotels"
 * @param string     $roleCode HA role code (language independent)
 * @param array      $profile  ['name' => …, 'email' => …, 'phone' => …] — display
only, not signed
 */
function sharptalk_identity(string $userSn, ?array $hotel, string $roleCode, array
$profile, string $secret): array {
    $attributes = ['role' => $roleCode];
    if ($hotel !== null) {
        $attributes['hotel_id']    = $hotel['sn'];
        $attributes['hotel_name']  = $hotel['name'];
        $attributes['province_sn'] = $hotel['provinceSn'];
    }
    $clean = [];
    foreach ($attributes as $k => $v) {
        $c = sharptalk_clean($v);
        if ($c !== '') { $clean[$k] = $c; }          // empty values are left out o
f signing AND payload
    }
    ksort($clean, SORT_STRING);
    $userId = 'ha:' . sharptalk_clean($userSn);
    $lines = [$userId];
    foreach ($clean as $k => $v) { $lines[] = $k . '=' . $v; }
    $hash = hash_hmac('sha256', implode("\n", $lines), $secret);

    $package = ['userId' => $userId, 'hash' => $hash];
    foreach (['name', 'email', 'phone'] as $f) {     // unsigned; drop when empty
        $c = sharptalk_clean($profile[$f] ?? '');
        if ($c !== '') { $package[$f] = $c; }
    }
```

```
    if ($clean) { $package['attributes'] = $clean; }
    return $package;
}

// Controller (login session required):
// $staff = currentStaff(); $hotel = selectedHotelOrNull();
// return json(sharptalk_identity($staff->sn, $hotel ? ['sn' => $hotel->sn, 'name'
=> $hotel->name,
//         'provinceSn' => $hotel->provinceSn] : null, $staff->roleCode,
//         ['name' => $staff->fullName, 'email' => $staff->email, 'phone' => $staff-
>mobile],
//         getenv('SHARPTALK_EMBED_SECRET')));
```

5.2 HA FE

```
<script>
  window.SHARPTALK_WIDGET_CONFIG = {
    shop: "https://www.go2joy.vn",
    widgetUrl: SHARPTALK_ORIGIN + "/widget",        // B7: one setting per environm
ent
    agent: "hotel-partner",
    locale: (localStorage.getItem("lang") || "vi").toLowerCase()
    // no `user` block, no hash here (B1)
  };
  window.SharpTalk = window.SharpTalk || { q: [] };
  function sharptalkCall(method, arg) {
    if (typeof window.SharpTalk[method] === "function") window.SharpTalk[method](ar
g);
    else window.SharpTalk.q.push([method, arg]);
  }
</script>
<script src="SHARPTALK_ORIGIN/widget/embed.js" defer></script>
```

```
// B2 · B3: after sign-in, on reload, on hotel change (incl. "all hotels"), on lang
uage change
async function sharptalkSignIn() {
  const pkg = await haApi.get('/api/v1/staff/sharptalkIdentity'); // signed on the
server
  sharptalkCall('identify', pkg);                                  // UNCHANGED
}
// B4: sign-out button, session/token expiry, before another account signs in, othe
r tab signed out
function sharptalkSignOut() { sharptalkCall('logout'); }
window.addEventListener('storage', (e) => { if (e.key === 'profile' && !e.newValue)
sharptalkSignOut(); });
// B5
function sharptalkSetLocale(lang) { sharptalkCall('setLocale', String(lang || 'v
i').toLowerCase()); }
// optional log
window.addEventListener('message', (e) => {
```

```
  if (e.origin !== SHARPTALK_ORIGIN || e.data?.type !== 'ivy:event') return;
  if (e.data.event === 'identified') console.info('[SharpTalk] identify', e.data.ok
? 'OK' : 'rejected');
});
```

5.3 Package example (T2)

```
{
  "userId": "ha:1001",
  "hash": "fb670d6fbaac555b40b2f9896ec7de0604a1fad83aaf9e892f293f1d3be9faa7",
  "name": "G2J SUPPORT",
  "attributes": { "hotel_id": "1721", "hotel_name": "A In Hotel Del Luna", "provinc
e_sn": "1", "role": "RECEPTIONIST" }
}
```

5.4 Test vectors (v1.4)

The test secret is `test-secret-do-not-use` (not a real secret). The hashes below were computed separately by **the SharpTalk server verification code** and by **the 5.1 PHP sample (PHP 8.3 + intl)**, and they match. Role codes and `userSn` are example values.

#	Input	Expected hash
T1	`ha:1001` · no attributes	`244f6fa4c1c48d077d54bd632b0897042fa413e36d55921517c462e2af51e618`
T2	`ha:1001` · hotel_id= `1721` , hotel_name= `A In Hotel Del Luna` , province_sn= `1` , role= `RECEPTIONIST`	`fb670d6fbaac555b40b2f9896ec7de0604a1fad83aaf9e892f293f1d3be9faa7`
T3	`ha:2002` · hotel_id= `2001` , hotel_name= `Huế Garden Hotel` , province_sn= `32` , role= `OWNER`	`0df74c490621c521a673b7bbd6014e6e3141a086d04c7c2f8de3699d31e581d7`
T4	`ha:3003` · hotel_id= `3001` , hotel_name= `Hạ Long Bay Hotel` , province_sn= `15` , role= `RECEPTIONIST`	`7e5cf87cdd3dc38574ec8f3f9fe6671e9650806d28f4741a55bb508f53b77ea9`
T5	`ha:1001` · role= `MANAGER` (**All hotels**)	`d57dc894632e52a7ca62d97f4cbec924eb33982dff23f57058b7299d24f57f20`
T6	Same values as T3, entered messy: hotel_id= `" 2001"` (leading space), hotel_name= `" Hu" + "ế" + " Garden Hotel "` (decomposed ế · double space · NBSP), province_sn= `32` (number type) → sign after 5.1 normalization	**Must equal T3** `0df74c490621c521a673b7bbd6014e6e3141a086d04c7c2f8de3699d31e581d7`

- Run: `docker run --rm -v "$PWD/scripts/go2joy/php":/w php:8.3-cli sh -c 'apt-get update -qq && apt-get install -y -qq libicu-dev && docker-php-ext-install intl && php /w/vectors.php'`

- Resulting province/region (values SharpTalk saves): T2 Hồ Chí Minh · `south`, T3 Thừa Thiên Huế · `north_center`, T4 Quảng Ninh · `north`, T5 none (default assignee).
- Do not use the v1.0 ~ v1.3 vectors (only T1 is the same).

5.5 SharpTalk responses

Result	Meaning	Action
200	Verified. Session is "verified", business customer and contact linked	—
401 · E5048	Signature mismatch: string·secret·environment mismatch, unknown key, over 120 chars or line break, NFC/whitespace difference, non-numeric `province_sn`. **On staging, until the temporary mode ends (§ 7), a mismatch on a session that is not yet verified is accepted as "Unverified" (200) instead**	Compare with section 4, rerun 5.4
400	Format error: empty or malformed email, `userId` over 120 chars, `name` over 200, `phone` over 40. The whole sign-in fails	Drop empty or invalid fields
404	No chat session	Widget handles it automatically. Reload

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6. Security

- Keep the secret only in HA server environment variables. Staging and production secrets differ (SharpTalk sends them over a secure channel).
- Anyone holding a signed package can chat as that staff member, see that staff member's earlier conversations, and get answers that use the hotel's previous conversation. Do not log it, and do not skip sign-out handling (B4).
- The AI provider receives the staff display name, hotel name, province, region and role (to greet and answer in context). Email, phone and the user id are not sent.
- Whether to add a signature expiry (`ts`) is decided in work request WR-03 (if added, it ships together as v1.5).
- Shared PCs (front desk): automatic sign-out and account switching in B4 are especially important.

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7. SharpTalk status

Item	Status
v1.4 support (`province_sn` → province name/region, v1.3 in parallel)	**Done on staging (10/24)**
Go2Joy province list, 80 entries (indexed by number)	Loaded on staging · production loaded at install
Temporary unsigned sign-in (staging only, for UAT)	Ends **as soon as a verified identify is seen on HA UAT** (A and B done), **before** the joint UAT — so the 401 rows of § 8 can pass. Until then a wrong signature is also accepted (as "Unverified")
Role code display names	After the A7 code table is received

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8. UAT checklist (WR-05) — the only UAT list

Used by HA and SharpTalk together (work request § 6.1 points here). Run it **after SharpTalk ends the temporary unsigned mode** (§ 7).

Check	Where	Expected
Widget while not signed in	HA UAT	Guest, consent notice shown (same as now)
HA sign-in → widget	HA UAT · console live chat	identify **200** with `"authenticated": true` and `consentState: "granted"`, no consent notice, staff name in greeting, console shows "verified"
Business customer	Console › Customers	1 hotel (name · province · region, calculated by SharpTalk — filled only by a signed sign-in), staff and role code under contacts, room types as that hotel's products
HA language vi ↔ en	HA UAT	Signed values unchanged → same conversation continues, only the reply language changes
Change hotel selection	HA UAT · console › Customers	Business customer and conversation switch to the new hotel
"All hotels" → connect to agent	HA UAT · console › Alarm center	200, no hotel, default assignee (work request WR-13)
Connect to agent from hotels in the 4 regions present in the feed (north · north_center · south_center · south) × consult category bs / cs	HA UAT · console › Agent handoff · Alarm center	Team × label per the console rules (work request WR-10), assignee shown in the alarm center
Skip the consult category	HA UAT · console › Alarm center	Default assignee (WR-13)
Assignment check	Console › Settings › Agent handoff	No "0 active" warning

Check	Where	Expected
Hotel name with many tone marks (incl. decomposed input)	HA UAT · console › Customers	200
Staff with empty email	HA UAT · console › Customers	200 without the email field
Sign-out · session expiry · another account signs in	HA UAT	Guest (or the new account), previous staff conversations not visible
Change one signed value without re-signing	HA UAT · console › Work log	401 E5048, `ok: false`, guest. `name` · `email` · `phone` are not signed, so changing them still gives 200
Add a key outside the list (e.g. `district`) and sign	HA UAT · console › Work log	401 (unknown keys are left out of the signature calculation)